

Behavioural Research in Climate Policy: A Report on *Misperceived Social Norms and Willingness to Act Against Climate Change*

Theo Osterhaus (Matriculation No. 7443693)
Faculty of Management, Economics and Social Sciences
University of Cologne

Seminar: Climate and Human Behavior (Spring 2026)
Instructor: Rastislav Rehák

It addresses behavioural research in climate policy, using the assigned focal paper as a starting point and anchor. The report summarises the main research question, methods, and findings of the focal paper, situates it within the broader course themes and literature on behavioural climate policy, and discusses its policy relevance and regulatory context. The total length of the report does not exceed 2500 words (excluding the bibliography), in line with the course requirements.

Contents

1	Summary of the Focal Paper	2
1.1	Motivation and Research Question	2
1.2	Methods and Identification Strategy	2
1.3	Main Results and Conclusions	2
2	Positioning within Course Themes and Literature	3
2.1	Central Behavioural Mechanisms	3
2.2	Relation to Behavioural Climate Policy Literature	3
3	Policy Relevance and Regulatory Context	4
3.1	Policy Domain and Current Landscape	4
3.2	Implementation Status and Practical Feasibility	5
3.3	Complementarity with Existing Instruments	6
4	Critical Assessment and Future Research	6
5	Conclusion	7

1 Summary of the Focal Paper

1.1 Motivation and Research Question

Given the threat of Global Warming, consolidated by the five IPCC 2023 reports, it is crucial to understand where Climate Change originates and design effective environmental policies and international agreements to tackle it. Policymakers should have access to all the necessary information about human behaviour, including beliefs, attitudes, motivations, and social interaction, to design policies in such a way that they are accepted by the general public and institutions.

In this context, it is essential to understand what determines an individual's willingness to fight Global Warming. Andre et al. (2024b) addressed this issue by examining how individuals perceive social norms and how this affects their willingness to fight Global Warming. They focus on three dimensions of climate-relevant behaviour: (i) the individual pro-climate behaviour (measured by an incentivised donation decision), (ii) beliefs about others' behaviour and normative views and (iii) support for environmental policies. Based on the measured results, they were able to address the problem of misperceived social norms influencing climate action. They then wanted to find out whether correcting these misperceptions could increase individuals' support for environmental policies.

1.2 Methods and Identification Strategy

Andre et al. (2024b) measure individuals' willingness to fight Global Warming by conducting a quota-based survey of 8,000 participants, who are representative of the US adult population, via a donation decision of an incentivised task. In addition to that, they collect detailed information about economic preferences, moral foundations and baseline characteristics of the participants. The measurements were used for a preliminary descriptive analysis of the determinants of Climate Change behaviour and the misperceptions of social norms among 2,000 participants. A randomised controlled trial (RCT) was also conducted as a follow-up information experiment, in which two information treatments were used to identify a causal relation between perceived social norms as key drivers for Climate Change behaviour.

1.3 Main Results and Conclusions

Perceived social norms and economic preferences are statistically significant predictors of Climate Change behaviour. It was found that Americans systematically underestimate the prevalence of climate-related social norms. An information experiment showed that informing respondents about the actual prevalence reduced these misperceptions and encouraged individuals to adopt climate-friendly behaviour. This effect is especially large among participants who are sceptical about global warming, a group that is generally difficult to reach.

Key Message

To sum up Many people act not out of conviction, but because they mistakenly believe that others are doing nothing. Correcting misperceived social norms could trigger a positive feedback loop where learning about the existing support of climate norms encourages Americans to take visible action against Climate Change, which encourages others to follow suit.

In a Policy Letter, Andre et al. (2021) summarises their main findings, which are aimed at decision-makers in politics and government. A followup study done by Andre et al. (2024a) was

able to replicate these results in a worldwide context, making an information treatment a unique solution for prompting and scaling pro-climate behaviour: a simple, scalable and cost-effective intervention.

2 Positioning within Course Themes and Literature

2.1 Central Behavioural Mechanisms

To fully contextualise the findings of Andre et al. (2024b), it is crucial to understand the underlying behavioural mechanisms that drive pro-environmental decision-making. Climate decisions are rarely made in a vacuum; rather, individuals are heavily embedded in social contexts. As Ganguli and Mengel 2026 demonstrate, social influence and conditional cooperation are foundational drivers of carbon dioxide mitigation efforts. People calibrate their own climate-friendly actions based on the observed or assumed behaviour of their peers.

This conditional nature of climate action is deeply intertwined with concerns about personal and social identity. Sherif and Simon 2025 highlight the trade-offs between impact, inspiration, and image in pro-environmental behaviours. They suggest that individuals are motivated to maintain a positive social image. Consequently, when the perceived normative baseline shifts—i.e., when individuals realise that a broad majority actually cares about Climate Change—acting sustainably becomes a prerequisite for maintaining a positive social standing. Correcting pluralistic ignorance, as done in the focal paper, essentially sets a new, pro-climate baseline for what is considered socially desirable.

Furthermore, it is important to distinguish social norms from conventional behavioural interventions. Gravert and Olsson Collentine 2021 explicitly contrast simple nudges with norms and monetary incentives, showing that nudges alone are often insufficient for long-term behavioural shifts such as habit formation in public transport. Instead, habit formation relies heavily on what is perceived as the societal norm. By addressing the fundamental misperception of these norms, Andre et al. 2024b tap into a distinct and powerful mechanism that goes beyond traditional nudging, providing a structural foundation for long-term behavioural change and policy acceptance.

Identify the central behavioural mechanism or mechanisms studied in the focal paper. Examples include social norms, bounded rationality, present bias, loss aversion, fairness concerns, moral identity, motivated reasoning, or identity-based polarisation. Relate these mechanisms to concepts discussed in the seminar, especially in the introductory lectures and in other students' presentations. Explain whether the focal paper reinforces, refines, or challenges commonly held views about the role of these mechanisms in climate-related behaviour. If relevant, relate the mechanism to broader frameworks such as the BUCKET model of core social goals or other integrative theories of motivation in climate behaviour.

2.2 Relation to Behavioural Climate Policy Literature

The focal paper makes a crucial contribution to the literature on the political economy of climate policy. Recent studies have shown that traditional economic tools, such as carbon pricing, often face severe public resistance, and that even structural solutions like tax rebates have limited impact on increasing public support Mildenberger et al. 2022. Consequently, scholars argue that effective carbon taxes inherently require flanking by 'green nudges' (Gravert and Shreedhar 2022). Andre et al. (2024b) extend this literature by identifying a specific, highly potent nudge: the correction of pluralistic ignorance. By proving that social influence (Ganguli and Mengel 2026)

is heavily skewed by misperceived norms, the authors provide a scalable behavioural intervention to increase mass support for climate policies...

Situate the focal paper within the wider academic literature on behavioural climate policy. Identify a few closely related studies and explain how the focal paper complements or extends them. For instance, you may discuss whether the paper builds on earlier work on social norm interventions, carbon pricing acceptance, climate nudges, or bargaining in climate negotiations. Discuss whether the focal paper primarily contributes new empirical evidence, a novel theoretical perspective, a methodological innovation, or a combination of these. Explain what gap in the literature the paper fills and why this gap is important. If the paper challenges an established result or introduces a new behavioural mechanism into the climate policy discourse, make this explicit and discuss the implications.

3 Policy Relevance and Regulatory Context

3.1 Policy Domain and Current Landscape

It is only in climate diplomacy that reciprocity is largely set aside in favour of voluntary commitments.

The current climate policy landscape is defined by the fundamental challenge of global cooperation. Currently, most international climate goals are being missed because voluntary commitments fail to overcome the inherent free-rider problem of global warming. As Ockenfels (2023) highlights, climate protection must be viewed as a basic public good—akin to public infrastructure, streets, or security. To solve this, Ockenfels argues that international negotiations must shift away from purely voluntary pledges towards reciprocal, mutual obligations. This entails establishing a "climate club" of willing nations that agree on a minimal, reciprocal carbon price, stabilised through cross-border payments and sanctions, while simultaneously ensuring that green technologies become cheaper than fossil fuels. Regional efforts are necessary but ultimately insufficient without this international reciprocity.

While Ockenfels (2023) addresses the macro-institutional design of climate policy, these structural solutions heavily depend on the behavioural mechanisms explored by Andre et al. (2024b). The free-rider problem exists not only between nations but also among individuals. Just as states require reciprocal commitments to act decisively, individuals base their climate-friendly behaviour on the perceived actions and norms of their peers. Andre et al. (2024b) demonstrate that widespread "pluralistic ignorance"—where individuals systematically underestimate the climate-friendly norms of their fellow citizens—acts as a "behavioural market failure." This misperception artificially dampens public support for precisely the kind of robust pricing policies and international climate clubs that Ockenfels (2023) advocates.

Within Europe, this interplay between macro-regulation and micro-behaviour is highly relevant for the evolving regulatory landscape. The EU is currently preparing a new integrated framework for European climate resilience and risk management (European Commission 2026a), building upon the European Green Deal and the EU Adaptation Strategy (past evaluations, public consultation, and was first announced in the European Green Deal in 2019). This forthcoming framework aims to introduce concepts like "climate resilience by design" and clear risk ownership across sectors.

To successfully implement such comprehensive policies—which inherently require a mix of economic instruments (like the Emissions Trading System), strict regulations, and subsidies—policymakers must secure broad public backing. Traditional climate policy tools remain central, but they can be significantly enhanced by behavioural instruments. As derived from the focal

paper, correcting misperceived social norms serves as a low-cost, scalable tool to strengthen the "license to regulate." By systematically integrating norm-based information into national and communal campaigns, policymakers can align public perception with actual majorities, thereby increasing the political feasibility of both local adaptation measures and international pricing regimes.

Specifically, the focal paper directly speaks to the policy domain of social norm campaigns and public information disclosure policies. In practice, interventions relying on social comparisons have already proven effective in the energy sector. For instance, the 'Opower' programmes in the US and parts of Europe successfully reduced household energy consumption by including social comparisons in utility bills, showing consumers how their energy use compares to their neighbours. However, on a broader national or EU regulatory level, communication strategies still predominantly focus on the urgency of the crisis (e.g., the European Climate Pact's emphasis on scientific facts) rather than systematically correcting normative misperceptions. At the same time, the regulatory landscape is expanding its economic instruments, such as the introduction of the EU ETS 2, which will price emissions from buildings and road transport starting in 2027. Introducing such direct carbon pricing mechanisms significantly affects citizens' daily lives and faces high risks of political backlash.

Behavioural economics highlights additional drivers of this gap: present bias, limited attention, status quo bias, and the power of social norms. Social norms and reciprocity strongly shape climate-relevant behaviour; people are more willing to act when they believe others also contribute. Misperceptions of prevailing norms (pluralistic ignorance) can constitute a behavioural market failure that dampens climate action. Ethical governance requires transparency and accuracy; the goal is correcting false beliefs, not secretly manipulating preferences.

3.2 Implementation Status and Practical Feasibility

The European *climate resilience and risk management – integrated framework* is a forthcoming EU initiative pushed by European Commission (2026a) that aims to establish a more ambitious, comprehensive and coherent approach to climate resilience and preparedness across all Member States and at EU level.

There is also a Call for Action paper by European Commission 2025 and an Factual Summary by the European Commission 2026b.

Its core objective is to help prevent and prepare for the growing impacts of Climate Change by creating a balanced policy package that combines legislative and non-legislative measures, economic instruments, and information tools, with adoption planned for the second half of 2026. The framework builds on existing EU adaptation strategies and responds to calls from the European Parliament, the Council and the European Court of Auditors for stronger action on climate risks.

Key elements already announced include a common climate reference scenario for Europe based on a 3°C global warming trajectory, which will serve as a benchmark for EU policies and provide a harmonised methodology for climate risk assessment plans by Member States, sectors and private actors. The initiative is also expected to introduce the concepts of "climate resilience by design" and "risk ownership", meaning that resilience considerations will be systematically integrated into policies, investments and planning, and that responsibilities for action will be clearly allocated in key sectors such as public health, energy, transport and defence.

In addition, the framework will provide decision-support tools based on Copernicus data to help stakeholders identify climate impacts in their regions or sectors, and will pay particular attention to closing finance and insurance protection gaps, given that only about 25% of climate-related

losses in Europe are currently insured.

Assess whether the approach studied in the focal paper is already implemented in some form, is under active discussion in policy circles, or is largely unrealised in practice. Draw on policy documents, reports, or other evidence where appropriate, and provide specific examples of implementations or proposals. Discuss any institutional, political, or ethical constraints that may affect the feasibility of applying the behavioural mechanism in real-world policy.

Reflect on the scalability and cost-effectiveness of the approach. For example, norm-based information campaigns may be relatively low-cost and scalable, but their effectiveness may depend on local social context and baseline beliefs. Behaviourally informed reforms of carbon pricing may face political economy constraints even if they are efficient in principle. Use these considerations to assess the practical value of the insights provided by the focal paper.

3.3 Complementarity with Existing Instruments

Link to existing regulatory instruments: Regulatory information to support: CO pricing and emissions trading (acceptance, perceived fairness), regulatory legislation (e.g. building standards, bans), subsidy schemes (e.g. heat pumps, e-mobility).

Konkrete Einsatzfelder für normbasierte Kommunikation

Nationale Informationskampagnen, die tatsächliche Mehrheiten sichtbar machen („die meisten in Deutschland tun / befürworten X“) Kommunale Programme (Energieberatung, ÖPNV, Mülltrennung) mit lokalen Normbotschaften („in unserer Stadt...“). Integration in Standard-Kommunikation von Ministerien, Behörden, Energieversorgern.

Discuss how the behavioural approach studied in the focal paper complements, challenges, or refines existing climate policy instruments. Consider whether the mechanism can be combined with traditional tools such as carbon pricing, regulation, subsidies, or information campaigns to enhance their effectiveness or public acceptability. Alternatively, discuss whether the mechanism suggests a rethinking of existing instruments or highlights new trade-offs.

If relevant, relate the focal paper to broader strategies in behavioural climate policy, such as the use of framing to increase support for carbon pricing, the deployment of climate nudges to facilitate pro-environmental choices, or the design of institutions that account for bounded rationality and social preferences. Reflect on how policymakers could integrate the specific insights from the paper into a coherent policy mix.

4 Critical Assessment and Future Research

andre2024 (Diskussion/Limitations des Papers) *ECONtribute* $pB_0362022$ *Aus deinen Notizen : Hinweis auf transparente Kommunikation statt manipulativen Nudgings*

Forschungs- und Umsetzungsagenda

Bedarf an Feldexperimenten in „echter“ Policy-Umgebung (z.B. kommunale Kampagnen, nationale Initiativen). Kooperation von Wissenschaft, Ministerien und NGOs zur Erhebung repräsentativer Normdaten und zur Evaluation von Norm-Kampagnen.

Integration into national Climate Change legislation and strategies (e.g. communication and education components in Climate Change action plans). Particular potential lies with groups characterised by high levels of scepticism and low intrinsic willingness to act on Climate Change; these groups respond measurably to information on social norms. A tailored approach depending on political orientation, age, education, etc. (heterogeneity of effects).

Provide your own critical assessment of the focal paper, focusing on its strengths and limitations from a policy perspective. Consider issues such as external validity, robustness of results, representativeness of the sample, and the realism of the behavioural assumptions. Reflect on whether the experimental or theoretical setting captures key features of real-world climate policy contexts.

Suggest directions for future research that would help to address remaining uncertainties or to extend the insights of the focal paper. This may include studying different populations, exploring additional behavioural mechanisms, testing the intervention in different institutional settings, or combining the approach with other policy instruments. Emphasise how such research could contribute to more effective and politically feasible climate policy.

Policy Implications

Summarise here, in a few sentences, your main conclusion about the policy relevance of the behavioural mechanism studied in the focal paper. Highlight what policymakers should learn from this research and under which conditions its recommendations are most applicable.

5 Conclusion

In the concluding section, briefly recapitulate the main insights from your report. Restate the central contribution of the focal paper to behavioural research in climate policy and summarise how you have positioned it within the broader literature and course themes. Highlight the most important aspects of the policy and regulatory context that condition the practical relevance of the paper's findings.

Emphasise the key behavioural mechanism or mechanisms that emerge as particularly important for climate policy design, and reflect on the overarching lesson this holds for the integration of behavioural insights into climate governance. Finally, briefly state how your assessment of the focal paper has shaped your own understanding of the opportunities and limits of behavioural approaches in addressing Climate Change.

References

- Andre, Peter, Teodora Boneva, Felix Chopra, and Armin Falk (2021). *Bereit zum Klimaschutz? Soziale Normen sind entscheidend?* ECONtribute Policy Brief 022. Bonn: ECONtribute – Markets & Public Policy. URL: https://www.econtribute.de/RePEc/ajk/ajkpbs/ECONtribute_PB_022_2021.pdf.
- (2024a). “Globally representative evidence on the actual and perceived support for climate action”. In: *Nature Climate Change* 14.3, pp. 253–259. ISSN: 1758-6798. DOI: [10.1038/s41558-024-01925-3](https://doi.org/10.1038/s41558-024-01925-3). URL: <https://doi.org/10.1038/s41558-024-01925-3>.
 - (June 2024b). “Misperceived Social Norms and Willingness to Act Against Climate Change”. In: *The Review of Economics and Statistics*, pp. 1–46. ISSN: 0034-6535. DOI: [10.1162/rest_a_01468](https://doi.org/10.1162/rest_a_01468). eprint: https://direct.mit.edu/rest/article-pdf/doi/10.1162/rest_a_01468/2383833/rest_a_01468.pdf. URL: https://doi.org/10.1162/rest_a_01468.
- European Commission (2025). *Call for evidence on the European climate resilience and risk management – integrated framework*. European Commission, “Have Your Say” portal. Support for the development of the new integrated EU framework on climate resilience and risk management.
- (2026a). *European climate resilience and risk management – integrated framework*. European Commission, “Have Your Say” initiative. Initiative 14770, accessed via the Better Regulation portal. URL: https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/14770-European-climate-resilience-and-risk-management-integrated-framework_en.
 - (2026b). *Factual summary report of the public consultation on the European climate resilience and risk management – integrated framework*. European Commission. Summary of contributions to the open public consultation.
- Ganguli, Jayant Vivek and Friederike Mengel (2026). “Social influence and carbon dioxide mitigation”. In: *Journal of Public Economics* 253, p. 105558. ISSN: 0047-2727. DOI: [10.1016/j.jpubeco.2025.105558](https://doi.org/10.1016/j.jpubeco.2025.105558).
- Gravert, Christina and Linus Olsson Collentine (2021). “When nudges aren’t enough: Norms, incentives and habit formation in public transport usage”. In: *Journal of Economic Behavior & Organization* 190, pp. 1–14. ISSN: 0167-2681. DOI: <https://doi.org/10.1016/j.jebo.2021.07.012>. URL: <https://www.sciencedirect.com/science/article/pii/S0167268121003000>.
- Gravert, Christina and Ganga Shreedhar (2022). “Effective carbon taxes need green nudges”. English. In: *Nature climate change* 12. Funding Information: The activities of CEBI are financed by the Danish National Research Foundation, grant DNRF134., pp. 1073–1074. ISSN: 1758-678X. DOI: [10.1038/s41558-022-01515-1](https://doi.org/10.1038/s41558-022-01515-1).
- Intergovernmental Panel on Climate Change (2023). *Climate Change 2023: Impacts, Adaptation, and Vulnerability*. Working Group II Contribution to the Sixth Assessment Report. Summary for Policymakers, page 25. URL: <https://www.ipcc.ch/report/ar6/wg2/>.
- Mildenberger, Matto, Erick Lachapelle, Kathryn Harrison, and Isabelle Stadelmann-Steffen (Feb. 2022). “Limited impacts of carbon tax rebate programmes on public support for carbon pricing”. In: *Nature Climate Change* 12.2, pp. 141–147. ISSN: 1758-6798. DOI: [10.1038/s41558-021-01268-3](https://doi.org/10.1038/s41558-021-01268-3). URL: <https://doi.org/10.1038/s41558-021-01268-3>.
- Ockenfels, Axel (2023). *Woran scheitert die Klimapolitik bislang?* ECONtribute Policy Brief 052. Bonn: ECONtribute – Markets & Public Policy. URL: https://www.econtribute.de/RePEc/ajk/ajkpbs/ECONtribute_PB_052_2023.pdf.
- Sherif, Raisa and Sven A. Simon (2025). “Impact, inspiration, or image: On the trade-offs in pro-environmental behaviors”. In: *Journal of Environmental Economics and Management* 134, p. 103214. ISSN: 0095-0696. DOI: <https://doi.org/10.1016/j.jeem.2025.103214>. URL: <https://www.sciencedirect.com/science/article/pii/S0095069625000981>.